



The National University of Computer and
Emerging Sciences

Dimensionality Reduction

Machine Learning for Data Science

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Goals

- Review of Previous Lecture
- Today's Lecture
 - Dimensionality Reduction
 - Principal Component Analysis

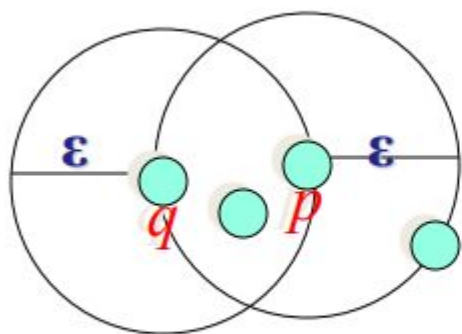
Review of Previous Lecture

Density-based Clustering

- All clustering methods use almost the same approach
 - Calculate similarities and then add to the cluster
- k -means is generally good, but:
 - Need to specify the number of clusters in beginning
 - K-Means may fail for different shapes.
 - Outliers can affect its performance ;
 - A slight change in data points *might* affect the clustering outcome.
- DBSCAN clustering can overcome these.

DBSCAN Clustering Algorithm

- The DBSCAN algorithm uses two parameters:
- **eps (ϵ)**: A **distance measure** that will be used to locate the points in the **neighborhood** of any point.
- **minPts**: The **minimum number of points** (a threshold) clustered together for a region to be **considered dense**.



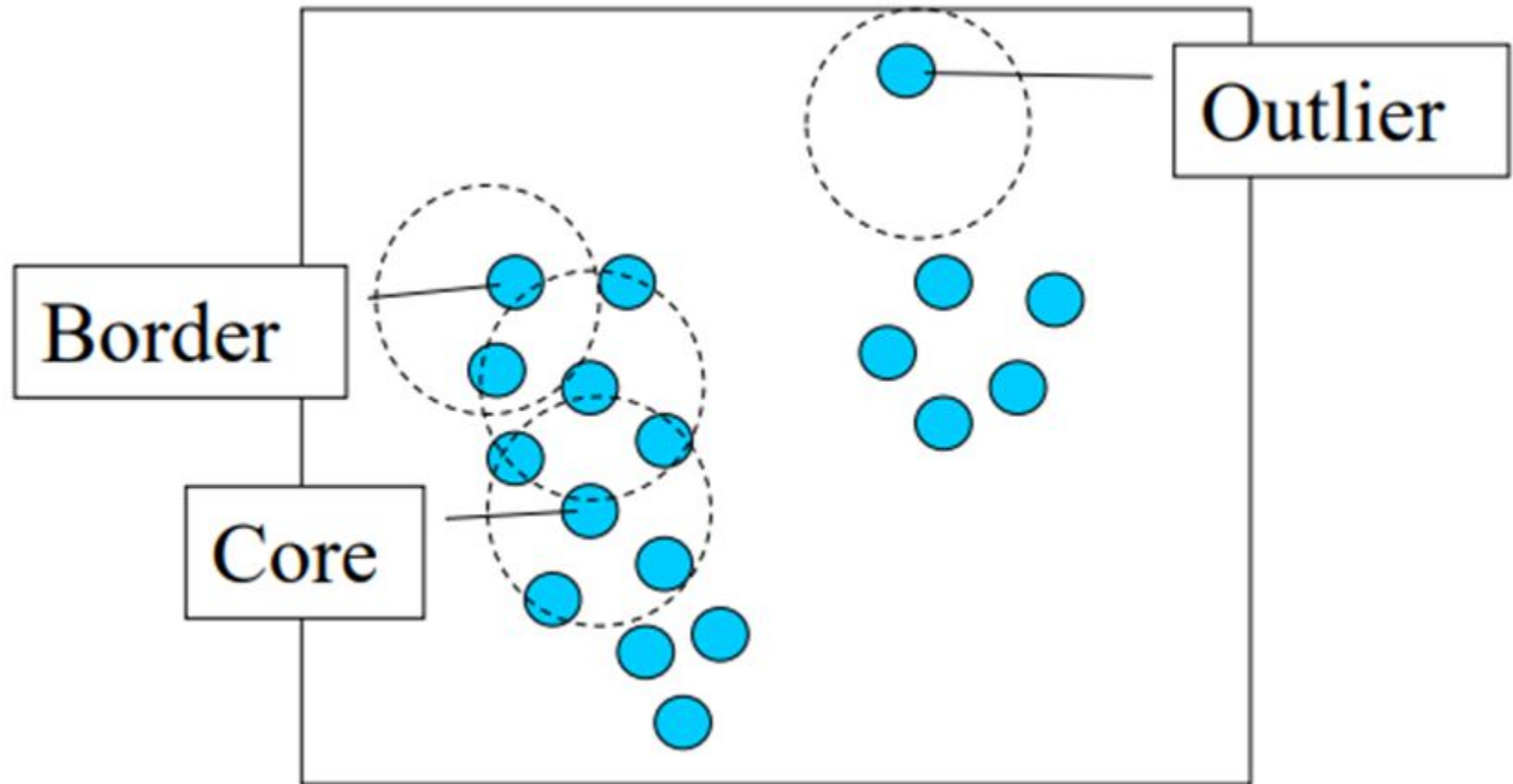
ϵ -Neighborhood of p

ϵ -Neighborhood of q

Density of p is “high” (MinPts = 4)

Density of q is “low” (MinPts = 4)

DBSCAN Clustering Algorithm



$$\epsilon = 2, \text{minPts} = 4$$

DBSCAN Clustering Algorithm

...are *not* added to clusters
and called **outliers**...



Today's Lecture

Dimensionality Reduction

- The **input data** fed to the model for **training** and then **decision making**.
 - Contain information
- *The **complexity of any classifier or regressor** depends on the **number of inputs**.*
 - *Dimensionality of input*
- *Time and space complexity*
- *Required number of **training examples** to train model.*

Dimensionality Reduction

- **Why dimensionality reduction?**
 - The **complexity** depends on the **number of input dimensions (d)**
 - Decreases the **complexity**
 - The **size** of the **data sample (N)**
 - Efficient **memory and computation**

Dimensionality Reduction

- **Why dimensionality reduction?**
- **Redundant features** must be removed.
 - BMI, Height, Weight
 - **Simpler** models are **more robust on small datasets**.
- **Fewer features can represent** the data in effective way (if possible)
 - *Hidden or latent factors*
- **Easier data visualization** with fewer dimensions

Dimensionality Reduction

- *How to perform dimensionality reduction?*
 - *Curse of Dimensionality*
- There are **two main** methods for reducing dimensionality:
 - Feature selection and feature extraction.
- **Feature selection:**
 - Select a **subset of features**.
 - Finding **k** of the **d dimensions** that give us the most information
- **Feature extraction:**
 - Finding a new set of k dimensions that are **combinations of the original d dimensions**.

Feature Selection

- Select on a best subset of features from original data.
- The **best subset** contains the **least number of dimensions**
 - Should not **affect the accuracy**.
- Discard the **remaining, unimportant dimensions**.
- Using a suitable **error function** to find best subset
- Testing all features may be hard (**polynomial time**).

Feature Selection

- There are two approaches for feature selection:
 - Forward Selection and Backward selection
- *Forward selection:*, Start with no feature and add them one by one
 - Add the feature that decreases the error
- *Backward selection:* Start with all features and remove them one by one.
 - Removing the one that decreases the error the most (or increases it only slightly).
- Checking the error with a validation dataset

Feature Extraction

- Component analysis is an unsupervised approach to finding the “right” features from the data.
- One leading method for feature extraction
 - Principal component analysis (PCA)
- PCA: Represent the d -dimensional data in a lower-dimensional space.
 - Represent same data in a lower dimensional space
 - Should best describe original data.
 - Should maintain the variations in the original data
- We standardize the data before applying PCA

Feature Extraction

- Mean

$$\mu_0 = \frac{1}{N} \sum_{i=1}^N \mathbf{x}_i$$

- Variance

$$\sigma^2 = \frac{\sum (x_i - \bar{x})^2}{N}$$

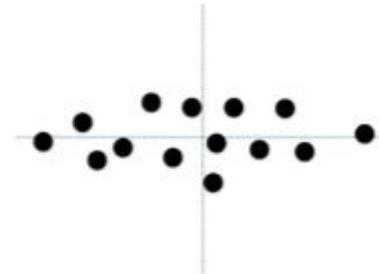
- Covariance

$$\text{Cov}(X, Y) = \frac{1}{n} \sum_{i=1}^n (x_i - \mu_x)(y_i - \mu_y)$$

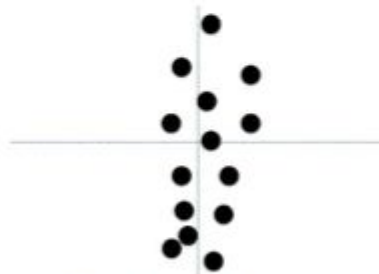
Basic Algebra



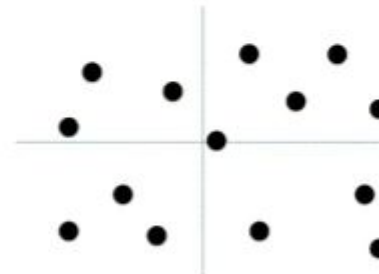
Small x-variance
Small y-variance



Large x-variance
Small y-variance

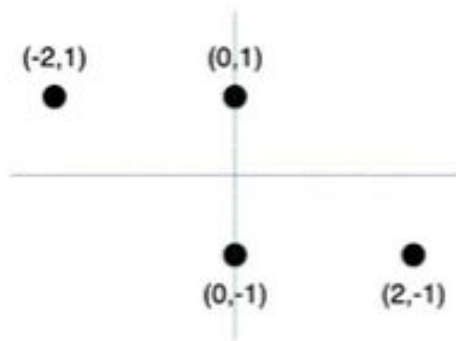


Small x-variance
Large y-variance



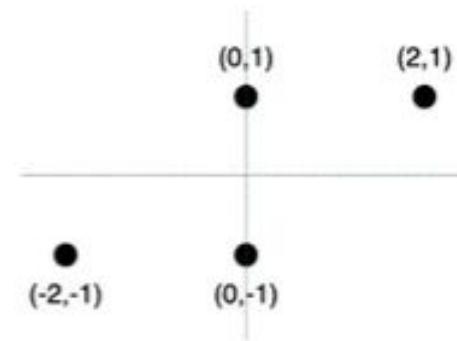
Large x-variance
Large y-variance

Basic Algebra



$$\text{var}(x) = \frac{1}{4}((-2)^2 + 0^2 + 0^2 + 2^2) = 2$$

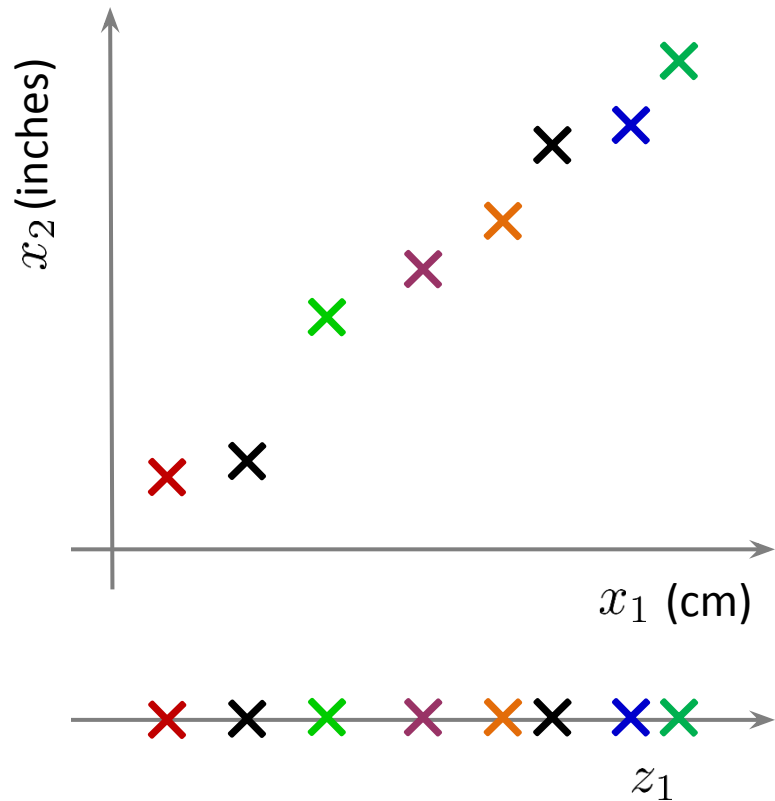
$$\text{var}(y) = \frac{1}{4}(1^2 + 1^2 + (-1)^2 + (-1)^2) = 1$$



$$\text{var}(x) = \frac{1}{4}((-2)^2 + 0^2 + 0^2 + 2^2) = 2$$

$$\text{var}(y) = \frac{1}{4}((-1)^2 + (-1)^2 + 1^2 + 1^2) = 1$$

Data Compression



Reduce data from
2D to 1D

$$x^{(1)} \rightarrow z^{(1)}$$

$$x^{(2)} \rightarrow z^{(2)}$$

$\vdots$

$$x^{(m)} \rightarrow z^{(m)}$$

PCA Algorithm

Goal: Find r -dim projection that best preserves variance

1. Compute mean vector μ and covariance matrix Σ of original points
2. Compute eigenvectors and eigenvalues of Σ
3. Select top r eigenvectors
4. Project points onto subspace spanned by them:

$$y = A(x - \mu)$$

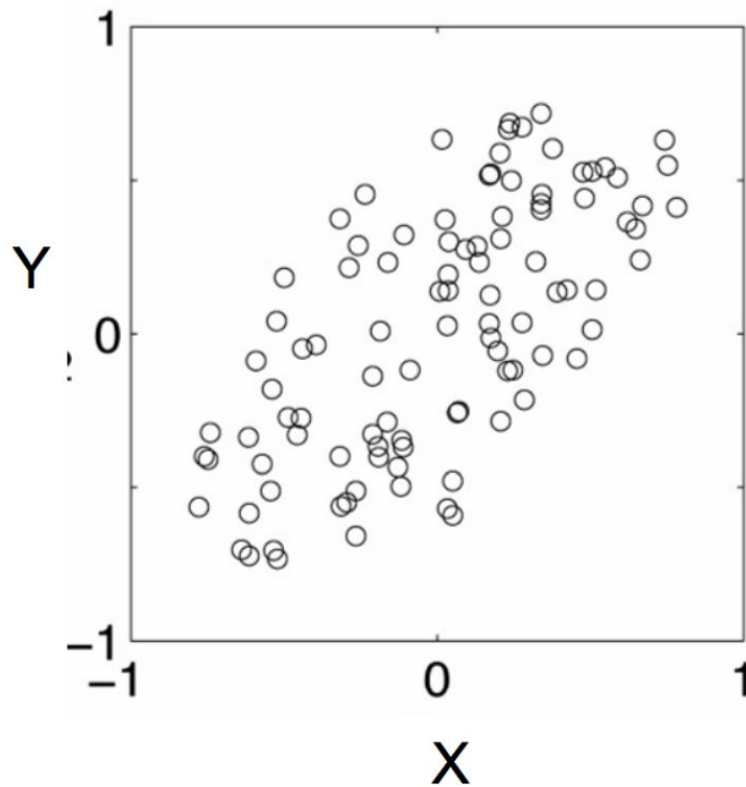
where y is the new point, x is the old one,
and the rows of A are the eigenvectors

Covariance

- **Variance and Covariance:**
 - Measure of the “spread” of a set of points around their center of mass(mean)
- **Variance:**
 - Measure of the deviation from the mean for points in one dimension
- **Covariance:**
 - Measure of how much each of the dimensions vary from the mean with **respect to each other**

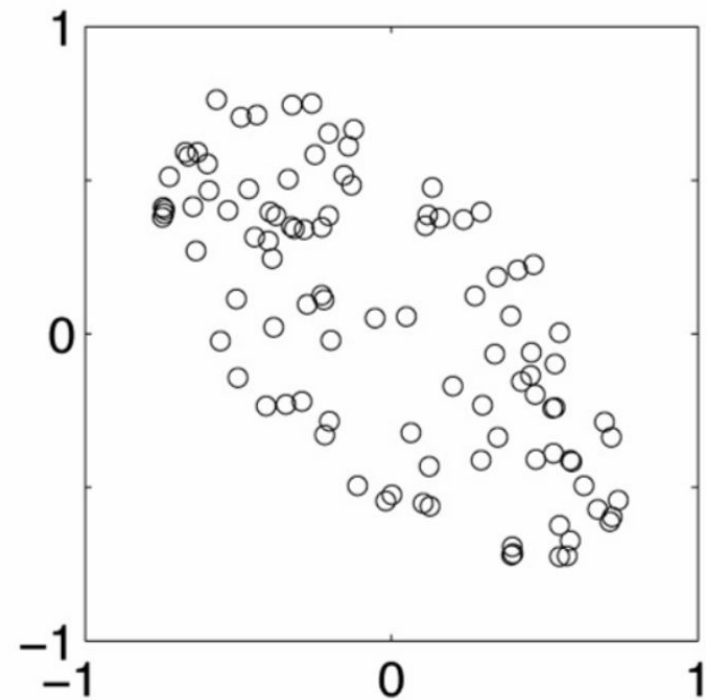
- Covariance is **measured between two dimensions**
- Covariance sees if there is a relation between two dimensions
- Covariance between one dimension is the variance

positive covariance



Positive: Both dimensions increase or decrease together

negative covariance



Negative: While one increase the other decrease

Eigenvector and Eigenvalue

- Let A be an $n \times n$ matrix
- An ***eigenvector*** of A is a ***nonzero vector*** v in $\mathbb{R}^n$ such that

$$Av = \lambda v$$

- Where **λ** is a scalar value.
- If $Av = \lambda v$ for $v \neq 0$, we say that λ is the ***eigenvalue*** ***for*** v , and that v is an ***eigenvector*** ***for*** λ .

Eigenvector and Eigenvalue

Consider the matrix

$$A = \begin{pmatrix} 2 & 2 \\ -4 & 8 \end{pmatrix} \quad \text{and vectors} \quad v = \begin{pmatrix} 1 \\ 1 \end{pmatrix} \quad w = \begin{pmatrix} 2 \\ 1 \end{pmatrix}.$$

Which are eigenvectors? What are their eigenvalues?

Solution

We have

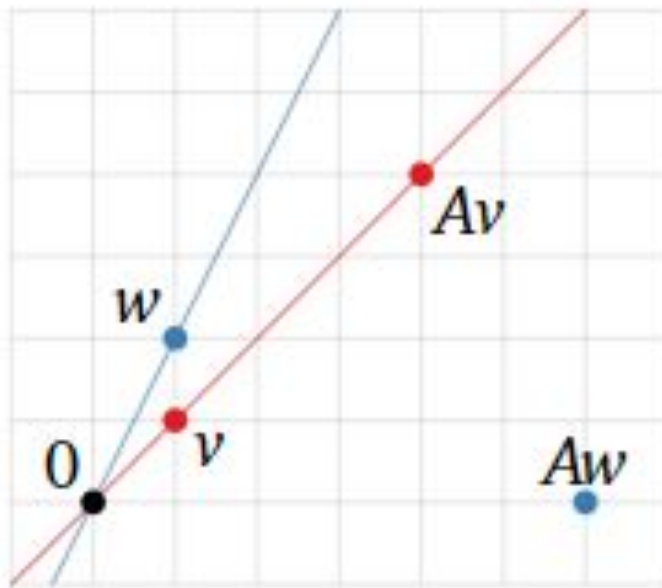
$$Av = \begin{pmatrix} 2 & 2 \\ -4 & 8 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 4 \\ 4 \end{pmatrix} = 4v.$$

Hence, v is an eigenvector of A , with eigenvalue $\lambda = 4$. On the other hand,

$$Aw = \begin{pmatrix} 2 & 2 \\ -4 & 8 \end{pmatrix} \begin{pmatrix} 2 \\ 1 \end{pmatrix} = \begin{pmatrix} 6 \\ 0 \end{pmatrix}.$$

which is not a scalar multiple of w . Hence, w is not an eigenvector of A .

Eigenvector and Eigenvalue



v is an eigenvector

w is not an eigenvector

Eigenvector and Eigenvalue

$$Ax = \lambda x$$

- A: Square Matrix
- λ : Eigenvector or characteristic vector
- X: Eigenvalue or characteristic value

Example

Show $x = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$ is an eigenvector for $A = \begin{bmatrix} 2 & -4 \\ 3 & -6 \end{bmatrix}$

$$\text{Solution : } Ax = \begin{bmatrix} 2 & -4 \\ 3 & -6 \end{bmatrix} \begin{bmatrix} 2 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\text{But for } \lambda = 0, \lambda x = 0 \begin{bmatrix} 2 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

Thus, x is an eigenvector of A , and $\lambda = 0$ is an eigenvalue.

Eigenvector and Eigenvalue

$$Ax = \lambda x \quad \rightarrow \quad \begin{aligned} Ax - \lambda x &= 0 \\ (A - \lambda I)x &= 0 \end{aligned}$$

x will be an eigenvector of A if and only if B does not have an inverse, or equivalently $\det(B)=0$:

$$\det(A - \lambda I) = 0$$

Eigenvector and Eigenvalue

• Example : Find the eigenvalues of

$$A = \begin{bmatrix} -6 & 3 \\ 4 & 5 \end{bmatrix}$$

$$|A - \lambda I| = 0$$

$$\left| \begin{bmatrix} -6 & 3 \\ 4 & 5 \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \right| = 0$$

$$\begin{vmatrix} -6 - \lambda & 3 \\ 4 & 5 - \lambda \end{vmatrix} = 0$$

$$(-6 - \lambda)(5 - \lambda) - 3 \times 4 = 0$$

$$\lambda^2 + \lambda - 42 = 0$$

$$\lambda = -7 \text{ or } 6$$

Eigenvector and Eigenvalue

$$Av = \lambda v$$

Put in the values we know:

$$\begin{bmatrix} -6 & 3 \\ 4 & 5 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = 6 \begin{bmatrix} x \\ y \end{bmatrix}$$

After multiplying we get these two equations:

$$-6x + 3y = 6x$$

$$4x + 5y = 6y$$

Bringing all to left hand side:

$$-12x + 3y = 0$$

$$4x - 1y = 0$$

Eigenvector and Eigenvalue

Either equation reveals that $\mathbf{y} = 4\mathbf{x}$, so the **eigenvector** is any non-zero multiple of this:

$$\begin{bmatrix} 1 \\ 4 \end{bmatrix}$$

$$\begin{bmatrix} -6 & 3 \\ 4 & 5 \end{bmatrix} \begin{bmatrix} 1 \\ 4 \end{bmatrix} = \begin{bmatrix} -6 \times 1 + 3 \times 4 \\ 4 \times 1 + 5 \times 4 \end{bmatrix} = \begin{bmatrix} 6 \\ 24 \end{bmatrix}$$

... and also ...

$$6 \begin{bmatrix} 1 \\ 4 \end{bmatrix} = \begin{bmatrix} 6 \\ 24 \end{bmatrix}$$

So $A\mathbf{v} = \lambda\mathbf{v}$

Principal Component Analysis

Input: $\mathbf{x} \in \mathbb{R}^D: \mathcal{D} = \{\mathbf{x}_1, \dots, \mathbf{x}_N\}$

Set of basis vectors: $\mathbf{u}_1, \dots, \mathbf{u}_K$

Summarize a D dimensional vector $\mathbf{x}$ with K dimensional feature vector $h(\mathbf{x})$

$$h(\mathbf{x}) = \begin{bmatrix} \mathbf{u}_1 \cdot \mathbf{x} \\ \mathbf{u}_2 \cdot \mathbf{x} \\ \dots \\ \mathbf{u}_K \cdot \mathbf{x} \end{bmatrix}$$

Principal Component Analysis

$$\mathbf{U} = [\mathbf{u}_1, \dots, \mathbf{u}_K]$$

Basis vectors are orthonormal

$$\mathbf{u}_i^T \mathbf{u}_j = 0$$

$$||\mathbf{u}_j|| = 1$$

New data representation $h(\mathbf{x})$

$$z_j = \mathbf{u}_j \cdot \mathbf{x}$$

$$h(\mathbf{x}) = [z_1, \dots, z_K]^T$$

Principal Component Analysis

$$\mathbf{U} = [\mathbf{u}_1, \dots, \mathbf{u}_K]$$

New data representation $h(\mathbf{x})$

$$h(\mathbf{x}) = \mathbf{U}^T (\mathbf{x} - \mu_0)$$

Empirical mean of the data

$$\mu_0 = \frac{1}{N} \sum_{i=1}^N \mathbf{x}_i$$

References

- Numerical Examples
 - <https://medium.com/analytics-vidhya/understanding-principle-component-analysis-pca-step-by-step-e7a4bb4031d9>
- For various Mathematical Visualization
 - <https://www.3blue1brown.com/>

Thank You 😊